

Commit ac35ccb

Browse files

 ActuallyNull committed on Aug 11

Model V.1.2, Spreadsheet updated, changed model architecture, added roc/auc curves, batch_size 64->100

 main 1 parent [0d30bde](#) commit ac35ccb 

 Filter files...

 model.py

 1 file changed +21 -22 lines changed

 Search within code

model.py

+21 -22

...

3 3

from keras.callbacks import EarlyStopping, ReduceLRonPlateau

4 4

from data_preperation import create_train_val_test_splits

5 5

from sklearn.metrics import confusion_matrix, classification_report

6

+ import matplotlib.pyplot as plt

7

+ from sklearn.metrics import RocCurveDisplay

8

+ from sklearn.preprocessing import label_binarize

6 9

7 10

X_train, y_train, X_val, y_val, X_test, y_test = create_train_val_test_splits()

8 11

...

...

11 14

ReduceLRonPlateau(monitor='val_loss', factor=0.2, patience=5, min_lr=1e-6, verbose=1)

12 15

]

13 16

14 -

15 17

model = tf.keras.Sequential([

16 -

layers.Conv1D(32, kernel_size=5),

17 -

layers.Conv1D(32, kernel_size=5),

18 -

layers.MaxPool1D(pool_size=2),

19 -

layers.Conv1D(64, kernel_size=5),

20 -

layers.Conv1D(64, kernel_size=5),

21 -

layers.MaxPool1D(pool_size=2),

22 -

layers.Conv1D(128, kernel_size=5),

23 -

layers.Conv1D(128, kernel_size=5),

24 -

layers.Conv1D(128, kernel_size=5),

25 -

layers.MaxPool1D(pool_size=2),

26 -

layers.Conv1D(256, kernel_size=5),

27 -

layers.Conv1D(256, kernel_size=5),

28 -

layers.Conv1D(256, kernel_size=5),

29 -

layers.MaxPool1D(pool_size=2),

30 -

layers.Conv1D(256, kernel_size=5),

31 -

layers.Conv1D(256, kernel_size=5),

32 -

layers.Conv1D(256, kernel_size=5),

33 -

layers.MaxPool1D(pool_size=2),

18 +

layers.Conv1D(60, 1),

19 +

layers.Conv1D(50, 1),

20 +

layers.Conv1D(40, 1),

34 21

layers.GlobalAveragePooling1D(),

35 -

layers.Dense(128, activation='relu'),

22 +

layers.Dense(20, activation='relu'),

36 23

layers.Dense(3, activation='softmax')

37 24

])

38 25

...

...

43 30

model.fit(X_train,

44 31

y_train,

45 32

epochs=100,

46 -

batch_size=64,

33 +

batch_size=100,

47 34

callbacks=callback,

48 35

)

49 36

...

...

59 46

print(confusion_matrix(y_test, y_pred_classes))

60 47

61 48

print("Classification Report:")

62 -

print(classification_report(y_test, y_pred_classes))

49 +

print(classification_report(y_test, y_pred_classes))

50 +

51 +

y_test_binary = label_binarize(y_test, classes=[0,1,2])

52 +

53 +

for i in range(y_pred.shape[1]):

54 +

RocCurveDisplay.from_predictions(y_test_binary[:,i], y_pred[:,i], name=f"Class {i}")

55 +

56 +

plt.plot([0,1], [0,1], 'k--')

57 +

plt.xlabel("FPR (False Positive Rate)")

58 +

plt.ylabel("TPR (True Positive Rate)")

59 +

plt.title("OvR ROC Curve")

60 +

plt.legend()

61 +

plt.show()

Comments 0



Please [sign in](#) to comment.